

As angular movements come under mechanical property and finally we restrain results in the form of digitalised data, these are labelled as MEMS sensors.

Another functionality of these sensors is that these act as temperature sensors. These are advantageous over other sensors and are helpful for developers who need both the sensors, since these have dual functionality, as temperature gyroscopes. Such sensors have many applications; these can be used in GPS trackers, digital compasses and as gravity sensors.

Force sensors. Force sensors are the best examples of MEMS technology. These sensors convert the change in force applied at the sensing area into changed electrical signal. These sensors are also called force sensing resistors (FSRs). These are used in small weighing machines and robotics applications such as artificial robotic hands, among others.

Humidity sensors. These sensors measure the surrounding air and, with calibrated-equivalent digitalised signals, provide atmospheric temperatures. These are based on the principle of absorption of gas in surrounding air, which in the absence of gravity can bend the microscopic cantilever, causing a change in the resistor doped on it, which is sensed electrically. These are mainly used in places such as power plants, mines, dockyards and green houses.

Pressure sensors. These are the most intelligent sensors and are used for measuring applied pressure which gets converted to voltage. There are numerous applications of these sensors, like bending of material where pressure is created within the material. Hence, by using these sensors we can find the breaking point of a material (maximum pressure that can be handled by the material) for quality assurance. These sensors are used in medical applications such as blood pressure measurement, industrial measurements, robotics, measurement of pressure in valves and in automobile manufacturing industries.

Touch sensors

Modern computers and electronic devices are touch based. Touchscreen technology is increasingly gaining popularity as can be seen in case of ATMs, mobile phones, information kiosks and vending machines, to name a few. Touchscreen based systems allow easy navigation around GUI based environments. There are several uniquely different types of touchscreens; the five most common types are 5-wire (or 4-wire) resistive, surface capacitive, projected capaci-

tive, surface acoustic wave (SAW) and infra-red (IR).

Wire resistive. A resistive touchscreen monitor (Fig. 1) has a glass panel and a film screen, each covered with a thin metallic layer of indium-tin-oxide (ITO) separated by a narrow gap. When a user touches the screen, two metallic layers make contact, resulting in flow of current. The point of contact is detected by the change in voltage. Resistive touchscreen panels are generally more affordable but offer only 75 per cent clarity, and the layer can be damaged by sharp objects.

The primary difference between 4-wire and 5-wire touchscreen is, in case of 5-wire, one wire goes to the cover-sheet, which serves as the voltage probe for the screen.

Surface capacitive. In a surface capacitive touchscreen monitor (Fig. 2), a transparent electrode film is placed on top of a glass panel, which has a protective cover. When an exposed finger touches the monitor screen, it reacts to the static electrical capacity of the human body; some of the electrical charge transfers from the screen to the user. This decrease in capacitance is detected by sensors located at the four corners of the screen, allowing the controller to determine the touch point. Capacitive touchscreens can only be activated by the touch of human skin or a stylus holding an electrical charge.

Projected capacitive. Projected capacitive (Fig. 3) is similar to surface capacitive but

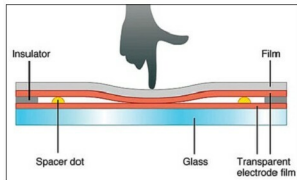


Fig. 1: Resistive capacitive

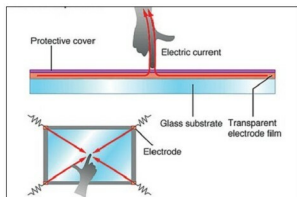


Fig. 2: Surface capacitive

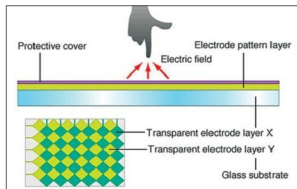


Fig. 3: Projected capacitive